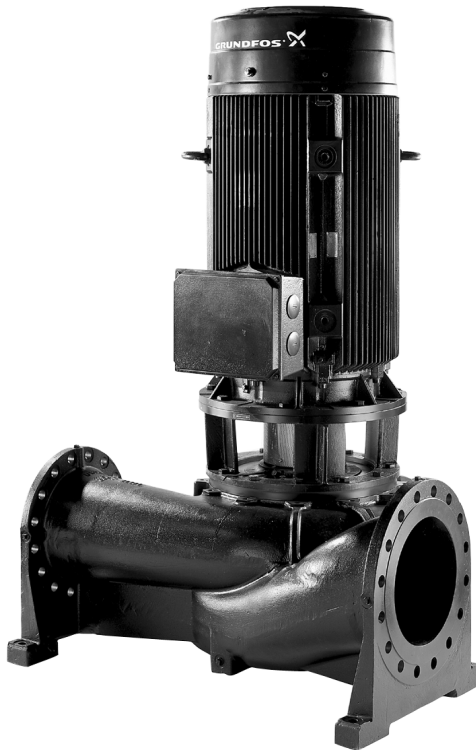


TP

Installation and operating instructions



Declaration of conformity

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the products TP, to which this declaration relates, are in conformity with these Council directives on the approximation of the laws of the EC member states:

- Machinery Directive (2006/42/EC).
Standards used: EN 809:1998 and EN 60204-1:2006.
- Ecodesign Directive (2009/125/EC).
Electric motors:
Commission Regulation No 640/2009.
Applies only to three-phase Grundfos motors marked IE2 or IE3. See motor nameplate.
Standard used: EN 60034-30:2009.
- Ecodesign Directive (2009/125/EC).
Water pumps:
Commission Regulation No 547/2012.
Applies only to water pumps marked with the minimum efficiency index MEI. See pump nameplate.
- ATEX Directive (94/9/EC) (applies only to products with the ATEX mark on the nameplate).
Standards used: EN 13463-1:2001 and EN 13463-5:2003.
(Declaration of conformity and installation and operating instructions of the motor are enclosed.)

This EC declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 96511031 1012).

BG: ЕС декларация за съответствие

Ние, фирма Grundfos, заявяваме с пълна отговорност, че продуктите TP, за които се отнася настоящата декларация, отговарят на следните указания на Съвета за уеднаквяване на правните разпоредби на държавите членки на ЕС:

- Директива за машините (2006/42/EC).
Приложени стандарти: EN 809:1998 и EN 60204-1:2006.
- Директива за екодизайн (2009/125/EC).
Електродвигатели:
Наредба No 640/2009 на Европейската комисия.
Отнася се само за трифазни електродвигатели на Grundfos, маркирани с IE2 или IE3. Вижте табелата с данни на двигателя.
Приложен стандарт: EN 60034-30:2009.
- Директива за екодизайн (2009/125/EC).
Водни помпи:
Наредба No 547/2012 на Европейската комисия.
Отнася се само за водни помпи, маркирани с минималният индекс за ефективност MEI.
Вижте табелата с данни на помпата.
- АТЕХ директива (94/9/EC) (отнася се само за продукти със символа АТЕХ върху табелата с данни).
Приложени стандарти: EN 13463-1:2001 и EN 13463-5:2003.
(Приложени са също и Декларацията за съответствие и инструкциите за монтаж и експлоатация на двигателя.)

Тази ЕС декларация за съответствие е валидна само когато е публикувана като част от инструкциите за монтаж и експлоатация на Grundfos (номер на публикацията 96511031 1012).

CZ: ES prohlášení o shodě

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobky TP, na něž se toto prohlášení vztahuje, jsou v souladu s ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství v oblastech:

- Směrnice pro strojní zařízení (2006/42/ES).
Použité normy: EN 809:1998 a EN 60204-1:2006.
- Směrnice o ekodesignu (2009/125/ES).
Elektrické motory:
Nařízení Komise č. 640/2009.
Platí pouze pro třífázové motory Grundfos označené IE2 nebo IE3. Viz typový štítek motoru.
Použitá norma: EN 60034-30:2009.
- Směrnice o ekodesignu (2009/125/ES).
Vodní čerpadla:
Nařízení Komise č. 547/2012.
Vztahuje se pouze na vodní čerpadla označená minimální účinností index MEI. Viz typový štítek čerpadla.
- Směrnice pro ATEX (94/9/ES) (týká se pouze výrobků nesoucích na typovém štítku značku ATEX).
Použité normy: EN 13463-1:2001 a EN 13463-5:2003.
(Prohlášení o konformitě a instalační a provozní předpisy motoru jsou přiloženy.)

Toto ES prohlášení o shodě je platné pouze tehdy, pokud je zveřejněno jako součást instalačních a provozních návodů Grundfos (publikace číslo 96511031 1012).

DK: EF-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produkterne TP som denne erklæring omhandler, er i overensstemmelse med disse af Rådets direktiver om indbyrdes tilnærmelse til EF-medlemsstaternes lovgivning:

- Maskindirektivet (2006/42/EF).
Anvendte standarder: EN 809:1998 og EN 60204-1:2006.
- Ecodesigndirektivet (2009/125/EF).
Elektriske motorer:
Kommissionens forordning nr. 640/2009.
Gælder kun 3-fasede Grundfos-motorer der er mærket IE2 eller IE3. Se motorens typeskilt.
Anvendt standard: EN 60034-30:2009.
- Ecodesigndirektivet (2009/125/EF).
Vandpumper:
Kommissionens forordning nr. 547/2012.
Gælder kun vandpumper der er mærket med mindsteeffektivitetsindekset MEI. Se pumpens typeskilt.
- ATEX-direktivet (94/9/EF)
(gælder kun for produkter med ATEX-mærkning på typeskiltet).
Anvendte standarder: EN 13463-1:2001 og EN 13463-5:2003.
(Motorens overensstemmelseserklæring og monterings- og driftsinstruktion er vedlagt.)

Denne EF-overensstemmelseserklæring er kun gyldig når den publiceres som en del af Grundfos-monterings- og driftsinstruktionen (publikations-nummer 96511031 1012).

DE: EG-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte TP, auf die sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmen:

- Maschinenrichtlinie (2006/42/EG).
Normen, die verwendet wurden: EN 809:1998 und EN 60204-1:2006.
- ErP-Richtlinie (2009/125/EG).
Elektromotoren:
Verordnung der Europäischen Kommission Nr. 640/2009.
Gilt nur für dreiphasige Motoren von Grundfos mit der Kennzeichnung IE2 bzw. IE3. Siehe Typenschild des Motors.
Norm, die verwendet wurde: EN 60034-30:2009.
- ErP-Richtlinie (2009/125/EG).
Wasserpumpen:
Verordnung der Europäischen Kommission Nr. 547/2012.
Gilt nur für Pumpen, für die der Mindesteffizienzindex (MEI) anzugeben ist. Siehe Typenschild der Pumpe.
- ATEX-Richtlinie (94/9/EG) (gilt nur für Produkte mit der ATEX-Kennzeichnung auf dem Typenschild).
Normen, die verwendet wurden: EN 13463-1:2001 und EN 13463-5:2003.
(Die Konformitätserklärung und Montage- und Betriebsanleitung vom Motor sind beigelegt.)

Diese EG-Konformitätserklärung gilt nur, wenn sie in Verbindung mit der Grundfos Montage- und Betriebsanleitung (Veröffentlichungsnummer 96511031 1012) veröffentlicht wird.

GR: Δήλωση συμμόρφωσης EC

Εμείς, η Grundfos, δηλώνουμε με αποκλειστικά δική μας ευθύνη ότι τα προϊόντα TP στα οποία αναφέρεται η παρούσα δήλωση, συμμορφώνονται με τις εξής Οδηγίες του Συμβουλίου περί προσέγγισης των νομοθεσιών των κρατών μελών της ΕΕ:

- Οδηγία για μηχανήματα (2006/42/EC).
Πρότυπα που χρησιμοποιήθηκαν: EN 809:1998 και EN 60204-1:2006.
- Οδηγία Οικολογικού Σχεδιασμού (2009/125/EC).
Ηλεκτρικοί κινητήρες:
Ρύθμιση πρώτης εκκίνησης Νο 640/2009.
Ισχύει μόνο σε τριφασικούς κινητήρες της Grundfos με σήμανση IE2 ή IE3. Βλέπε πινακίδα κινητήρα.
Πρότυπο που χρησιμοποιήθηκε: EN 60034-30:2009.
- Οδηγία Οικολογικού Σχεδιασμού (2009/125/EC).
Αντλίες νερού:
Ρύθμιση πρώτης εκκίνησης Νο 547/2012.
Ισχύει μόνο για αντλίες νερού που φέρουν τον ελάχιστο δείκτη απόδοσης MEI. Βλέπε πινακίδα αντλίας.
- Οδηγία ATEX (94/9/EC) (εφαρμόζεται μόνο σε προϊόντα με το σήμα ATEX στην πινακίδα τους).
Πρότυπα που χρησιμοποιήθηκαν: EN 13463-1:2001 και EN 13463-5:2003.
(Περιλαμβάνονται δήλωση συμμόρφωσης και οδηγίες εγκατάστασης και λειτουργίας του κινητήρα.)

Αυτή η δήλωση συμμόρφωσης EC ισχύει μόνον όταν συνοδεύει τις οδηγίες εγκατάστασης και λειτουργίας της Grundfos
(κωδικός εντύπου 96511031 1012).

ES: Declaración CE de conformidad

Nosotros, Grundfos, declaramos bajo nuestra entera responsabilidad que los productos TP, a los cuales se refiere esta declaración, están conformes con las Directivas del Consejo en la aproximación de las leyes de los Estados Miembros del EM:

- Directiva de Maquinaria (2006/42/CE).
Normas aplicadas: EN 809:1998 y EN 60204-1:2006.
- Directiva sobre diseño ecológico (2009/125/CE).
Motores eléctricos:
Reglamento de la Comisión N° 640/2009.
Válido sólo para motores trifásicos Grundfos pertenecientes a las categorías IE2 e IE3. Véase la placa de características del motor.
Norma aplicada: EN 60034-30:2009.
- Directiva sobre diseño ecológico (2009/125/CE).
Bombas de agua:
Reglamento de la Comisión N° 547/2012.
Aplicable únicamente a las bombas de agua marcadas con el índice de eficiencia mínima (IEM). Véase la placa de características de la bomba.
- Directiva ATEX (94/9/CE) (se refiere sólo a productos con la marca ATEX en la placa de características).
Normas aplicadas: EN 13463-1:2001 y EN 13463-5:2003.
(Se adjuntan la declaración de conformidad e instrucciones de instalación y funcionamiento del motor.)

Esta declaración CE de conformidad sólo es válida cuando se publique como parte de las instrucciones de instalación y funcionamiento de Grundfos (número de publicación 96511031 1012).

FR: Déclaration de conformité CE

Nous, Grundfos, déclarons sous notre seule responsabilité, que les produits TP, auxquels se réfère cette déclaration, sont conformes aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous:

- Directive Machines (2006/42/CE).
Normes utilisées: EN 809 :1998 et EN 60204-1 :2006.
- Directive sur l'éco-conception (2009/125/CE).
Moteurs électriques:
Règlement de la Commission N° 640/2009.
S'applique uniquement aux moteurs triphasés Grundfos marqués IE2 ou IE3. Voir plaque signalétique du moteur.
Norme utilisée: EN 60034-30 :2009.
- Directive sur l'éco-conception (2009/125/CE).
Pompes à eau:
Règlement de la Commission N° 547/2012.
S'applique uniquement aux pompes à eau marquées de l'indice de performance minimum IEM. Voir plaque signalétique de la pompe.
- Directive ATEX (94/9/CE) (s'applique uniquement aux produits avec norme ATEX citée sur la plaque signalétique).
Normes utilisées: EN 13463-1 :2001 et EN 13463-5 :2003.
(Déclaration de conformité et notice d'installation et fonctionnement du moteur incluses.)

Cette déclaration de conformité CE est uniquement valide lors de sa publication dans la notice d'installation et de fonctionnement Grundfos (numéro de publication 96511031 1012).

HR: EZ izjava o usklađenosti

Mi, Grundfos, izjavljujemo pod vlastitom odgovornošću da je proizvod TP, na koji se ova izjava odnosi, u skladu s direktivama ovog Vijeća o usklađivanju zakona država članica EU:

- Direktiva za strojeve (2006/42/EZ).
Korištene norme: EN 809:1998 i EN 60204-1:2006.
- Direktiva o ekološkoj izvedbi (2009/125/EZ).
Električni motori:
Uredba Komisije No 640/2009.
Odnosi se samo na trofazne Grundfos motore s oznakama IE2 ili IE3. Pogledajte natpisnu pločicu motora.
Korištena norma: EN 60034-30:2009.
- Direktiva o ekološkoj izvedbi (2009/125/EZ).
Crpke za vodu:
Uredba Komisije No 547/2012.
Odnosi se samo na crpke za vodu označene s indeksom minimalne učinkovitosti MEI. Pogledajte natpisnu pločicu crpke.
- ATEX uredba (94/9/EZ) (vrijedi samo za proizvode s ATEX-znakom na natpisnoj pločici).
Korištene norme: EN 13463-1:2001 i EN 13463-5:2003.
(Deklaracija o usklađenosti te motažne i pogonske upute priloženi su uz motor.)

Ova EZ izjava o sukladnosti važeća je jedino kada je izdana kao dio Grundfos montažnih i pogonskih uputa (broj izdanja 96511031 1012).

IT: Dichiarazione di conformità CE

Grundfos dichiara sotto la sua esclusiva responsabilità che i prodotti TP, ai quali si riferisce questa dichiarazione, sono conformi alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri CE:

- Direttiva Macchine (2006/42/CE).
Norme applicate: EN 809:1998 e EN 60204-1:2006.
- Direttiva Ecodesign (2009/125/CE).
Motori elettrici:
Regolamento CE n. 640/2009.
Applicabile solo ai motori trifase Grundfos contrassegnati IE2 o IE3. Vedere la targhetta di identificazione del motore.
Norma applicata: EN 60034-30:2009.
- Direttiva Ecodesign (2009/125/CE).
Pompe per acqua:
Regolamento CE n. 547/2012.
Applicabile solo a pompe per acqua con l'indice di efficienza minimo MEI. Vedere la targhetta di identificazione della pompa.
- Direttiva ATEX (94/9/CE) (si applica solo ai prodotti che riportano la sigla ATEX sull'etichetta).
Norme applicate: EN 13463-1:2001 e EN 13463-5:2003.
(In allegato la dichiarazione di conformità e il manuale di installazione e funzionamento.)

Questa dichiarazione di conformità CE è valida solo quando pubblicata come parte delle istruzioni di installazione e funzionamento Grundfos (pubblicazione numero 96511031 1012).

HU: EK megfeleléségi nyilatkozat

Mi, a Grundfos, egyedüli felelősséggel kijelentjük, hogy a TP termékek, amelyekre jelen nyilatkozik vonatkozik, megfelelnek az Európai Unió tagállamainak jogi irányelveit összehangoló tanács alábbi előírásainak:

- Gépek (2006/42/EK).
Alkalmazott szabványok: EN 809:1998 és EN 60204-1:2006.
- Környezetbarát tervezésre vonatkozó irányelv (2009/125/EK).
Villamos motorok:
Az Európai Bizottság 640/2009. számú rendelete.
Csak az IE2 vagy IE3 jelzésű háromfázisú Grundfos motorokra vonatkozik. Lásd a motor adattábláját.
Alkalmazott szabvány: EN 60034-30:2009.
- Környezetbarát tervezésre vonatkozó irányelv (2009/125/EK).
Víz szivattyúk:
Az Európai Bizottság 547/2012. számú rendelete.
Csak a MEI minimum hatásfok index-el jelölt víz szivattyúkra vonatkozik. Lásd a szivattyú adattábláját.
- ATEX Direktíva (94/9/EK) (csak az ATEX jelzéssel ellátott termékekre vonatkozik).
Alkalmazott szabványok: EN 13463-1:2001 és EN 13463-5:2003.
(A motor kezelési utasítása és megfeleléségi nyilatkozata mellékelve.)

Ez az EK megfeleléségi nyilatkozat kizárólag akkor érvényes, ha Grundfos telepítési és üzemeltetési utasítás (kiadvány szám 96511031 1012) részeként kerül kiadásra.

NL: EC**overeenkomstigheidsverklaring**

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat de producten TP waarop deze verklaring betrekking heeft, in overeenstemming zijn met de Richtlijnen van de Raad in zake de onderlinge aanpassing van de wetgeving van de EG Lidstaten betreffende:

- Machine Richtlijn (2006/42/EC).
Gebruikte normen: EN 809:1998 en EN 60204-1:2006.
- Ecodesign Richtlijn (2009/125/EC).
Elektromotoren:
Verordening (EG) Nr. 640/2009 van de Commissie.
Geldt alleen voor de driefase elektromotoren van Grundfos, aangeduid met IE2 of IE3. Zie het typeplaatje van de motor.
Gebruikte norm: EN 60034-30:2009.
- Ecodesign Richtlijn (2009/125/EC).
Waterpompen:
Verordening (EG) Nr. 547/2012 van de Commissie.
Is alleen van toepassing op waterpompen die gekenmerkt worden door de minimale efficiëntie index MEI. Zie het typeplaatje van de pomp.
- ATEX Richtlijn (94/9/EC) (alleen van toepassing voor producten met de ATEX markering op de typeplaat).
Gebruikte normen: EN 13463-1:2001 en EN 13463-5:2003.
(Overeenkomstigheidsverklaring is ingesloten in de installatie- en bedieningsinstructies van de motor.)

Deze EC overeenkomstigheidsverklaring is alleen geldig wanneer deze gepubliceerd is als onderdeel van de Grundfos installatie- en bedieningsinstructies (publicatienummer 96511031 1012).

PL: Deklaracja zgodności WE

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze wyroby TP, których deklaracja niniejsza dotyczy, są zgodne z następującymi wytycznymi Rady d/s ujednolicenia przepisów prawnych krajów członkowskich WE:

- Dyrektywa Maszynowa (2006/42/WE).
Zastosowane normy: EN 809:1998 i EN 60204-1:2006.
- Dyrektywa Ekoprojektowa (2009/125/WE).
Silniki elektryczne:
Rozporządzenie komisji nr 640/2009.
Dotyczy tylko trójfazowych silników firmy Grundfos z oznaczeniami IE2 lub IE3. Patrz tabliczka znamionowa silnika.
Zastosowana norma: EN 60034-30:2009.
- Dyrektywa Ekoprojektowa (2009/125/WE).
Pompy do wody:
Rozporządzenie komisji nr 547/2012.
Dotyczy tylko pomp do tłoczenia wody z minimalnym indeksem sprawności MEI. Patrz tabliczka znamionowa pompy.
- Dyrektywa ATEX (94/9/WE) (dotyczy tylko wyrobów ze znakiem ATEX na tabliczce znamionowej).
Zastosowane normy: EN 13463-1:2001 i EN 13463-5:2003.
(Deklaracja zgodności oraz instrukcja montażu i eksploatacji silnika są załączone.)

Deklaracja zgodności WE jest ważna tylko i wyłącznie wtedy kiedy jest opublikowana przez firmę Grundfos i umieszczona w instrukcji montażu i eksploatacji (numer publikacji 96511031 1012).

PT: Declaração de conformidade CE

A Grundfos declara sob sua única responsabilidade que os produtos TP, aos quais diz respeito esta declaração, estão em conformidade com as seguintes Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da CE:

- Directiva Máquinas (2006/42/CE).
Normas utilizadas: EN 809:1998 e EN 60204-1:2006.
- Directiva de Conceção Ecológica (2009/125/CE).
Motores eléctricos:
Regulamento da Comissão No 640/2009.
Aplica-se apenas a motores trifásicos Grundfos assinalados como IE2 ou IE3. Ver a chapa de características do motor.
Norma utilizada: EN 60034-30:2009.
- Directiva de Conceção Ecológica (2009/125/CE).
Bombas de água:
Regulamento da Comissão No 547/2012.
Aplica-se apenas a bombas de água registadas com o índice de eficiência mínimo MEI. Ver a chapa de características da bomba.
- Directiva ATEX (94/9/CE) (apenas aplicável a produtos com a inscrição ATEX gravada na chapa de características).
Normas utilizadas: EN 13463-1:2001 e EN 13463-5:2003.
(Em anexo encontra a Declaração de conformidade e instruções de instalação e funcionamento do motor.)

Esta declaração de conformidade CE é apenas válida quando publicada como parte das instruções de instalação e funcionamento Grundfos (número de publicação 96511031 1012).

RU: Декларация о соответствии ЕС

Мы, компания Grundfos, со всей ответственностью заявляем, что изделия TP, к которым относятся настоящая декларация, соответствуют следующим Директивам Совета Евросоюза об унификации законодательных предписаний стран-членов ЕС:

- Механические устройства (2006/42/EC).
Применявшиеся стандарты: EN 809:1998 и EN 60204-1:2006.
- Директива по экологическому проектированию энергопотребляющей продукции (2009/125/EC).
Электродвигатели:
Регламент Комиссии ЕС № 640/2009.
Применяется только к трехфазным электродвигателям Grundfos, обозначенным IE2 или IE3. См. шильдик с техническими данными двигателя.
Применявшийся стандарт: EN 60034-30:2009.
- Директива по экологическому проектированию энергопотребляющей продукции (2009/125/EC).
Насосы для перекачивания воды:
Регламент Комиссии ЕС № 547/2012.
Применимо только к насосам для перекачивания воды, промаркированным показателем минимальной эффективности MEI. См. фирменную табличку насоса.
- Директива ATEX (94/9/EC) (действительно только для изделий с маркировкой ATEX на фирменной табличке с техническими данными).
Применявшиеся стандарты: EN 13463-1:2001 и EN 13463-5:2003.
(Заявление о соответствии и руководство по монтажу и эксплуатации электродвигателя прилагаются.)

Данная декларация о соответствии ЕС имеет силу только в случае публикации в составе инструкции по монтажу и эксплуатации на продукцию производства компании Grundfos (номер публикации 96511031 1012).

RO: Declarație de conformitate CE

Noi, Grundfos, declarăm pe propria răspundere că produsele TP, la care se referă această declarație, sunt în conformitate cu aceste Directive de Consiliu asupra armonizării legilor Statelor Membre CE:

- Directiva Utilaje (2006/42/CE).
Standarde utilizate: EN 809:1998 și EN 60204-1:2006.
- Directiva Ecodesign (2009/125/CE).
Motoare electrice:
Regulamentul Comisiei nr. 640/2009.
Se aplică numai motoarelor trifazate Grundfos cu marca IE2 sau IE3. Vezi plăcuța de identificare a motorului.
Standard utilizat: EN 60034-30:2009.
- Directiva Ecodesign (2009/125/CE).
Pompe de apă:
Regulamentul Comisiei nr. 547/2012.
Se aplică numai pompelor de apă cu marca de eficiența minimă index MEI. Vezi plăcuța de identificare a pompei.
- Directiva ATEX (94/9/CE) (se aplică numai la produsele cu marca ATEX pe plăcuța de identificare).
Standarde utilizate: EN 13463-1:2001 și EN 13463-5:2003.
(Declarația de conformitate și instrucțiunile de instalare și utilizare ale motorului sunt incluse.)

Această declarație de conformitate CE este valabilă numai când este publicată ca parte a instrucțiunilor Grundfos de instalare și utilizare (număr publicație 96511031 1012).

SK: Prehlásenie o konformite EÚ

My firma Grundfos prehlasujeme na svoju plnú zodpovednosť, že výrobky TP, na ktoré sa toto prehlásenie vzťahuje, sú v súlade s ustanovením smernice Rady pre zblíženie právnych predpisov členských štátov Európskeho spoločenstva v oblastiach:

- Smernica pre strojové zariadenie (2006/42/ES).
Použité normy: EN 809:1998 a EN 60204-1:2006.
- Smernica o ekodizajne (2009/125/ES).
Elektromotory:
Nariadenie Komisie č. 640/2009.
Platné iba pre trojfázové motory Grundfos, označené ako IE2 alebo IE3. Viď typový štítok motora.
Použitá norma: EN 60034-30:2009.
- Smernica o ekodizajne (2009/125/ES).
Čerpadlá na vodu:
Nariadenie Komisie č. 547/2012.
Vzťahuje sa iba na čerpadlá pre vodu označené minimálnym indexom energetickej účinnosti MEI.
Viď typový štítok čerpadla.
- Smernica pre ATEX (94/9/ES) (týka sa iba výrobkov nesúcich na typovom štítku značku ATEX).
Použité normy: EN 13463-1:2001 a EN 13463-5:2003.
(Prehlásenie o konformite a montážny a prevádzkový návod motora sú priložené.)

Toto prehlásenie o konformite ES je platné iba vtedy, ak je zverejnené ako súčasť montážnych a prevádzkových pokynov Grundfos (publikácia číslo 96511031 1012).

SI: ES izjava o skladnosti

V Grundfosu s polno odgovornostjo izjavljamo, da so naši izdelki TP, na katere se ta izjava nanaša, v skladu z naslednjimi direktivami Sveta o približevanju zakonodaje za izenačevanje pravnih predpisov držav članic ES:

- Direktiva o strojih (2006/42/ES).
Uporabljeni normi: EN 809:1998 in EN 60204-1:2006.
- Eco-design direktiva (2009/125/ES).
Električni motorji:
Uredba Komisije št. 640/2009.
Se nanaša samo na trofazne motorje Grundfos z oznako IE2 ali IE3. Glejte tipsko ploščico motorja.
Uporabljena norma: EN 60034-30:2009.
- Eco-design direktiva (2009/125/ES).
Vodne črpalke:
Uredba Komisije št. 547/2012.
Velja le za vodne črpalke označene z indeksom minimalne učinkovitosti MEI. Glejte tipsko ploščico črpalke.
- ATEX direktiva (94/9/ES) (velja samo za izdelke z oznako ATEX na tipski ploščici).
Uporabljeni normi: EN 13463-1:2001 in EN 13463-5:2003.
(Izjava o ustreznosti ter navodila za montažo in obratovanje motorja sta priložena.)

ES izjava o skladnosti velja samo kadar je izdana kot del Grundfos instalacije in navodil delovanja (publikacija številka 96511031 1012).

RS: EC deklaracija o konformitetu

Mi, Grundfos, izjavljujemo pod vlastitom odgovornošću da je proizvod TP, na koji se ova izjava odnosi, u skladu sa direktivama Saveta za usklađivanje zakona država članica EU:

- Direktiva za mašine (2006/42/EC).
Korišćeni standardi: EN 809:1998 i EN 60204-1:2006.
- Direktiva o ekološkom projektovanju (2009/125/EC).
Električni motori:
Uredba komisije br. 640/2009.
Važi samo za trofazne Grundfos motore označene sa IE2 ili IE3. Pogledajte natpisnu pločicu motora.
Korišćen standard: EN 60034-30:2009.
- Direktiva o ekološkom projektovanju (2009/125/EC).
Pumpe za vodu:
Uredba Komisije br. 547/2012.
Odnosi se samo na pumpe za vodu označene sa indeksom minimalne efikasnosti MEI. Pogledajte natpisnu pločicu pumpe.
- ATEX direktiva (94/9/EC) (odnosi se samo na proizvode sa natpisom ATEX na natpisnoj pločici).
Korišćeni standardi: EN 13463-1:2001 i EN 13463-5:2003.
(Deklaracija konformiteta i uputstva za instalaciju i rad motora su priloženi.)

Ova EC deklaracija o usaglašenosti važeća je jedino kada je izdata kao deo Grundfos uputstava za instalaciju i rad
(broj izdanja 96511031 1012).

FI: EY-**vaatimustenmukaisuusvakuutus**

Me, Grundfos, vakuutamme omalla vastuullamme, että tuotteet TP, joita tämä vakuutus koskee, ovat EY:n jäsenvaltioiden lainsäädännön yhdenmukaistamiseen tähtäävien Euroopan neuvoston direktiivien vaatimusten mukaisia seuraavasti:

- Konedirektiivi (2006/42/EY).
Sovellettavat standardit: EN 809:1998 ja EN 60204-1:2006.
- Ekologista suunnittelua koskeva direktiivi (2009/125/EY).
Sähkömootorit:
Komission asetus nro 640/2009.
Koskee vain Grundfosin IE2- tai IE3-merkittyjä 3-vaihemootoreita. Katso moottorin arvokilvestä.
Sovellettu standardi: EN 60034-30:2009.
- Ekologista suunnittelua koskeva direktiivi (2009/125/EY).
Vesipumput:
Komission asetus nro 547/2012.
Koskee vain vesipumppuja, jotka on merkitty minimihyötysuhdeindeksillä MEI. Katso pumpun tyyppikilvestä.
- ATEX-direktiivi (94/9/EY) (soveltuu vain tuotteisiin, joissa on ATEX-merkintä arvokilvestä).
Sovellettavat standardit: EN 13463-1:2001 ja EN 13463-5:2003.
(Moottorin vaatimustenmukaisuusvakuutus ja käyttöohjeet sisältyvät toimitukseen.)

Tämä EY-vaatimustenmukaisuusvakuutus on voimassa vain, kun se julkaistaan osana Grundfosin asennus- ja käyttöohjeita
(julkaisun numero 96511031 1012).

SE: EG-försäkran om överensstämmelse

Vi, Grundfos, försäkrar under ansvar att produkterna TP, som omfattas av denna försäkran, är i överensstämmelse med rådets direktiv om inbördes närmande till EU-medlemsstaternas lagstiftning, avseende:

- Maskindirektivet (2006/42/EG).
Tillämpade standarder: EN 809:1998 och EN 60204-1:2006.
- Ekodesigndirektivet (2009/125/EG).
Elektriska motorer:
Kommissionens förordning nr. 640/2009.
Gäller endast trefas Grundfos-motorer märkta med IE2 eller IE3. Se motorns typskylt.
Tillämpade standard: EN 60034-30:2009.
- Ekodesigndirektivet (2009/125/EG).
Vattenpumpar:
Kommissionens förordning nr. 547/2012.
Avser endast vattenpumpar markerade med min. effektivitetsindex (MEI). Se pumpens typskylt.
- ATEX-direktivet (94/9/EG) (endast för produkter med ATEX-märkning på typskylten).
Tillämpade standarder: EN 13463-1:2001 och EN 13463-5:2003.
(Försäkran om överensstämmelse samt monterings- och driftsinstruktion medföljer medlevererad motor.)

Denna EG-försäkran om överensstämmelse är endast giltig när den publiceras som en del av Grundfos monterings- och driftsinstruktion (publikation nummer 96511031 1012).

TR: EC uygunluk bildirgesi

Grundfos olarak bu beyannameye konu olan TP ürünlerinin, AB Üyesi Ülkelerin kanunlarını birbirine yaklaştırmaya üzerine Konsey Direktifleriyle uyumlu olduğunun yalnızca bizim sorumluluğumuz altında olduğunu beyan ederiz:

- Makineler Yönetmeliği (2006/42/EC).
Kullanılan standartlar: EN 809:1998 ve EN 60204-1:2006.
- Çevreye duyarlı tasarım (Ecodesign) Direktifi (2009/125/EC).
Elektrikli motorlar:
640/2009 sayılı Komisyon Yönetmeliği.
Sadece IE2 veya IE3 işaretli trifaze Grundfos motorlar için geçerlidir. Motor bilgi etiketine bakınız.
Kullanılan standart: EN 60034-30:2009.
- Çevreye duyarlı tasarım (Ecodesign) Direktifi (2009/125/EC).
Devirdaim su pompaları:
547/2012 sayılı Komisyon Yönetmeliği.
Yalnızca Minimum Enerji Verimlilik Endeksine (MEI) dahil olan devirdaim su pompaları için geçerlidir. Pompanın bilgi etiketine bakın.
- ATEX Yönergesi (94/9/EC) (sadece etiketinde ATEX işareti bulunan ürünler için geçerlidir).
(Motorun uygunluk beyannamesi ve montaj ve kullanım bilgileri arkaya eklenmiştir.)

İşbu EC uygunluk bildirgesi, yalnızca Grundfos kurulum ve çalıştırma talimatlarının (basım numarası 96511031 1012) bir parçası olarak basıldığı takdirde geçerlilik kazanmaktadır.

Bjerringbro, 15th October 2012



Svend Aage Kaas
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Person authorised to compile technical file and empowered to sign the EC declaration of conformity.



Декларация о соответствии на территории РФ

Насосы центробежные вертикальные одноступенчатые ТР сертифицированы на соответствие требованиям Технического регламента о безопасности машин и оборудования
(Постановление правительства РФ от 15.09.2009 № 753).

Сертификат соответствия:

№ С-RU.АЯ56.В.04430, срок действия до 13.09.2017г.

№ С-DK.АЯ56.В.03740, срок действия до 27.05.2017г.

Изделия, произведенные в России, изготавливаются в соответствии с ТУ 3631-008-59379130-2006.

Истра, 15 октября 2012 г.

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English (GB) Installation and operating instructions

Original installation and operating instructions.

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Warning
Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document



Warning
If these safety instructions are not observed, it may result in personal injury.



Caution
If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Note
Notes or instructions that make the job easier and ensure safe operation.

2. General

These instructions apply to TP pumps fitted with Grundfos motors, type MMG. If the pump is fitted with a motor make other than Grundfos, please note that the motor data may differ from the data stated in these instructions.

3. Versions

The TP pumps described in these instructions are available in two versions:

- PN 10 (10 bar) and
- PN 25 (25 bar).

See nameplate.

4. Delivery and handling

4.1 Delivery

The pump is delivered from the factory in a carton with a wooden bottom, which is specially designed for transport by fork-lift truck or a similar vehicle.

Pumps in PN 10 version are supplied mounted on base plate.

On pumps in PN 25 version the pump flanges are designed to support the entire pump.

4.2 Handling



Warning
The lifting eyes fitted to the motor can be used for lifting the pump head (motor, motor stool and impeller). The lifting eyes must not be used for lifting the entire pump.

The pump should be lifted by means of nylon straps and shackles, see fig. 1.

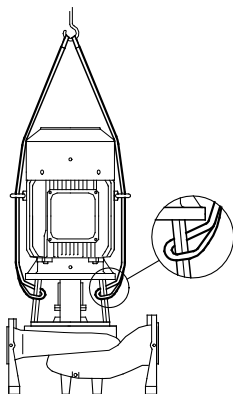


Fig. 1 TP

TM02 6991 2203

5. Applications

Grundfos single-stage in-line centrifugal pumps, type TP, are typically used for

- circulation in heating systems, including district heating
- circulation in ventilating and air-conditioning systems
- circulation and transfer in cooling systems
- distribution and pressure boosting in water supply systems
- circulation, transfer and pressure boosting in industrial systems.

5.1 Pumped liquids

Thin, clean, non-aggressive and non-explosive liquids, not containing solid particles or fibres. The liquid must not attack the pump materials chemically.

When pumping liquids with a density and/or viscosity higher than that of water, motors with correspondingly higher outputs must be used, if required.

The O-rings and the mechanical shaft seal chosen must be suitable for the liquid to be pumped.

Special shaft seals may be required if the pump is used for pumping treated water at temperatures above 80 °C containing additives to prevent system corrosion, calcareous deposits, etc., e.g. in heating and ventilating systems.

When fitted with the correct shaft seal, the pump can be used for pumping liquids at temperatures down to -25 °C.

6. Technical data

6.1 Ambient temperature

Maximum +40 °C.

6.2 Liquid temperature

See pump nameplate.

Depending on the cast-iron version and the pump application, the maximum liquid temperature may be limited by local regulations and laws.

6.3 Minimum inlet pressure

To ensure optimum pump operation, the inlet pressure (system pressure) must be adjusted correctly, see page 201.

6.4 Maximum inlet pressure

The actual inlet pressure + pressure when the pump is running against a closed valve must always be lower than the "maximum operating pressure", see pump nameplate.

6.5 Electrical data

See motor nameplate.

6.6 Sound level

See page 206.

7. Installation

7.1 Pump location

The pump should be sited in a dry, well ventilated, but frost-free position.



Warning

When pumping hot liquids, care should be taken to ensure that persons cannot accidentally come into contact with hot surfaces.

To ensure sufficient ventilation to the motor and pump, at least 0.5 metre clearance should be left above the pump.

Furthermore, it should be ensured that sufficient clearance is available to allow the use of lifting equipment when the pump/motor is to be serviced.

7.2 Pipework

Arrows on the pump housing show the direction of flow of liquid through the pump.

The pump is suitable for mounting in horizontal pipes.

Isolating valves should be fitted either side of the pump to avoid draining the system if the pump needs to be cleaned or repaired.

When installing the pipes, it must be ensured that the pump housing is not stressed by the pipework.

The suction and discharge pipes must be of an adequate size, taking the pump inlet pressure into account.

To avoid sediment build-up, do not fit the pump at the lowest point of the system.

Install the pipes so that air locks are avoided, especially on the suction side of the pump, see fig. 2.

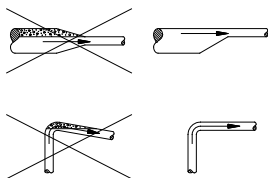


Fig. 2 Correct pipework on the suction side of the pump

TM00 2263 0195

7.3 Bypass

Caution

The pump is not allowed to run against a closed discharge valve as this will cause an increase in temperature/formation of steam in the pump which may cause damage to the pump.

If there is any danger of the pump running against a closed discharge valve, a minimum liquid flow through the pump should be ensured by connecting a bypass/a drain to the discharge pipe. The drain can for instance be connected to a tank. A minimum flow rate equal to 10 % of the flow rate at maximum efficiency is needed at all times.

Flow rate and head at maximum efficiency are stated on the pump nameplate.

7.4 Foundation

Grundfos recommends to install the pump on a concrete foundation which is heavy enough to provide permanent and rigid support to the entire pump. The foundation must be capable of absorbing any vibration, normal strain or shock. As a rule of thumb, the weight of the concrete foundation should be 1.5 times the weight of the pump.

Place the pump on the foundation and fasten it, see fig. 3.

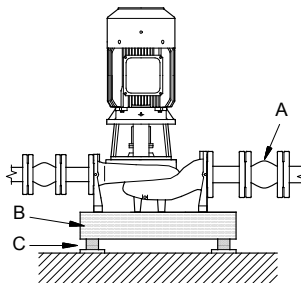


Fig. 3 Foundation for TP pump

A: Expansion joint

B: Concrete pedestal

C: Vibration damper

7.5 Vibration dampening

To prevent vibrations from being transmitted to the building and pipework, it is recommended to fit expansion joints and vibration dampers, see fig. 3.

7.5.1 Expansion joints

Expansion joints provide the following functions:

- Absorption of thermal expansion and contraction of pipework caused by variations in liquid temperature.
- Reduction of mechanical influences in connection with pressure surges in the pipework.
- Isolation of structure-borne noise in the pipework (only rubber bellows expansion joints).

Caution

Expansion joints must not be fitted to make up for inaccuracy in the pipework, e.g. centre displacement or misalignment of flanges.

The expansion joints should be fitted at a minimum distance of 1 - 1 1/2 x DN diameter from the pump, both on the suction and the discharge sides. This prevents turbulence in the joints, thus ensuring optimum suction conditions and minimum pressure drop on the discharge side.

At high water velocities (> 5 m/s), it is recommended to fit larger expansion joints matching the pipework.

TM02 6966 2003

7.5.2 Vibration dampers

To prevent vibrations from being transmitted to the building, it is recommended to isolate the pump foundation from buildings by means of vibration dampers. In order to select the right vibration damper, the following details are required:

- Forces transferred through the damper.
- Motor speed. In the case of speed control, this must also be taken into account.
- Desired dampening in % (recommended value 70 %).

The selection of vibration damper differs from installation to installation. A wrong damper may in certain cases increase the vibration level.

Vibration dampers should therefore be sized by the supplier.

If the pump is installed on a foundation with vibration dampers, expansion joints must be fitted on both sides of the pump. This is very important to ensure that the pump is not "hanging" from the flanges.

8. Electrical connections

The electrical connections should be carried out by an authorized electrician in accordance with local regulations.

Warning



Before removing the terminal box cover and before any removal/dismantling of the pump, make sure that the electricity supply has been switched off.

The pump must be connected to an external mains switch.

The operating voltage and frequency are marked on the nameplate. Make sure that the motor is suitable for the electricity supply on which it will be used.

The motor must be connected to a motor starter.

Motors of 3 kW and up incorporate thermistors (PTC). The thermistors are designed according to DIN 44082. The motor starter and the thermistors must be connected in series. Consequently, the motor will not be started until it has cooled to normal temperature.

The terminal box can be turned in 30 ° or 45 ° steps, depending on pump size.

Remove the bolts securing the motor to the pump.

Turn the motor to the required position. Replace and tighten the bolts.

The electrical connection should be carried out as shown in the diagram inside the terminal box cover.

Caution

Do not start the pump until it has been filled with liquid and vented.

8.1 Frequency converter operation

Grundfos motors:

All three-phase Grundfos motors from frame size 90 and up can be connected to a frequency converter.

The connection of a frequency converter will often have the effect that the motor insulation system is loaded more and that the motor will be more noisy than during normal operation. In addition, large motors are loaded by bearing currents caused by the frequency converter.

In the case of frequency converter operation, the following should be considered:

- In 2- and 4-pole motors of 110 kW and up and 6-pole motors of 75 kW and up, one of the motor bearings should be electrically isolated to prevent damaging currents from passing through the motor bearings.
- In the case of noise critical applications, the motor noise can be reduced by fitting a dU/dt filter between the motor and the frequency converter. In particularly noise critical applications, it is recommended to fit a sinusoidal filter.
- The length of the cable between motor and frequency converter affects the motor load. It should therefore be checked that the cable length meets the specifications laid down by the frequency converter supplier.
- For supply voltages between 500 and 690 V, either a dU/dt filter should be fitted to reduce voltage peaks or a motor with reinforced insulation should be used.
- For supply voltages of 690 V, a motor with reinforced insulation should be used and a dU/dt filter should be fitted.

Other motor makes than Grundfos:

Please contact Grundfos or the motor manufacturer.

9. Start-up

Caution

Do not start the pump until it has been filled with liquid and vented.

9.1 Priming

Closed systems or open systems where the liquid level is above the pump inlet:

1. Close the discharge isolating valve and loosen the air vent screw in the motor stool, see fig. 4.

Warning

Pay attention to the direction of the vent hole and take care to ensure that the escaping water does not cause injury to persons or damage to the motor or other components. In hot-water installations, special attention should be paid to the risk of injury caused by scalding hot water.

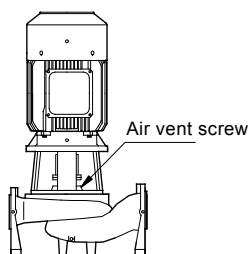


Fig. 4 Air vent screw

2. Slowly open the isolating valve in the suction pipe until a steady stream of liquid runs out of the vent hole.
3. Tighten the air vent screw and completely open the isolating valve(s).

Open systems where the liquid level is below the pump inlet:

The suction pipe and the pump must be filled with liquid and vented before the pump is started.

1. Close the discharge isolating valve and open the isolating valve in the suction pipe.
2. Loosen the air vent screw.
3. Remove the plug from one of the pump flanges, depending on the pump location.
4. Pour liquid through the priming port until the suction pipe and the pump are completely filled with liquid.
5. Replace the plug and tighten securely.
6. Tighten the air vent screw.

The suction pipe can to some extent be filled with liquid and vented before it is connected to the pump. A priming device can also be installed before the pump.

9.2 Checking the direction of rotation

Do not start the pump to check the direction of rotation until it has been filled with liquid.

The correct direction of rotation is indicated by arrows on the pump.

9.3 Starting

1. Before starting the pump, completely open the isolating valve on the suction side of the pump and leave the discharge isolating valve almost closed.
2. Start the pump.
3. Vent the pump during starting by loosening the air vent screw in the motor stool until a steady stream of liquid runs out of the vent hole, see fig. 4.

Warning

Pay attention to the direction of the vent hole and take care to ensure that the escaping water does not cause injury to persons or damage to the motor or other components. In hot-water installations, special attention should be paid to the risk of injury caused by scalding hot water.



4. When the piping system has been filled with liquid, slowly open the discharge isolating valve until it is completely open.

If the pumps are fitted with motors with outputs selected on the basis of a specific maximum flow rate, the motors may be overloaded if the differential pressure is lower than anticipated.

Caution

Measure motor current consumption and compare the result with the nominal current stated on the motor nameplate. In the event of motor overload, throttle the discharge isolating valve until the motor is no longer overloaded.

It is advisable always to check the motor current consumption during starting.

9.4 Frequency of starts and stops

The pump should not start more than 20 times per hour.

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10. Maintenance



Warning

Before starting work on the pump, make sure that the electricity supply has been switched off and that it cannot be accidentally switched on.

10.1 Pump

The pump is maintenance-free.

If the pump is to be drained for a long period of inactivity, inject a few drops of silicone oil on the shaft between the motor stool and the coupling. This will prevent the shaft seal faces from sticking.

10.2 Motor

The motor should be checked at regular intervals. It is important to keep the motor clean in order to ensure adequate ventilation. If the pump is installed in a dusty environment, the pump must be cleaned and checked regularly.

Lubrication:

The bearings of motors up to 11 kW are greased for life and require no lubrication.

The bearings of motors of 11 kW and up must be greased in accordance with the indications on the motor nameplate.

The motor should be lubricated with a lithium-based grease meeting the following specifications:

- NLGI grade 2 or 3.
- Viscosity of basic oil: 70 to 150 cSt at +40 °C (~ +104 °F).
- Temperature range: -30 °C (~ -22 °F) to +140 °C (~ +284 °F) during continuous operation.

11. Frost protection

Pumps which are not being used during periods of frost should be drained to avoid damage.

Drain the pump by loosening the air vent screw in the motor stool and by removing the drain plug from the pump housing, see fig. 5.



Warning

Care must be taken to ensure that the escaping water does not cause injury to persons or damage to the motor or other components.

In hot water installations, special attention should be paid to the risk of injury caused by scalding hot water.

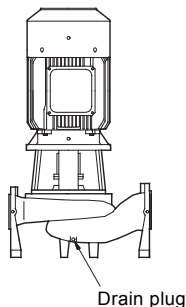


Fig. 5 Drain plug

Do not tighten the air vent screw and replace the drain plug until the pump is to be used again.

TM02 6968 2003

12. Fault finding chart



Warning

Before removing the terminal box cover and before any removal/dismantling of the pump, make sure that the electricity supply has been switched off and that it cannot be accidentally switched on.

Fault	Cause
1. Motor does not run when started.	a) Supply failure. b) Fuses blown. c) Motor starter overload has tripped out. d) Main contacts in motor starter are not making contact or the coil is faulty. e) Control circuit fuses are defective. f) Motor is defective.
2. Motor starter overload trips out immediately when supply is switched on.	a) One fuse is blown. b) Contacts in motor starter overload are faulty. c) Cable connection is loose or faulty. d) Motor winding is defective. e) Pump mechanically blocked. f) Overload setting too low.
3. Motor starter overload trips out occasionally.	a) Overload setting too low. b) Supply voltage periodically too low or too high. c) Differential pressure across pump too low, see section 9.3 <i>Starting</i> .
4. Motor starter has not tripped out but the pump does not run.	a) Supply failure. b) Fuses blown. c) Main contacts in motor starter are not making contact or the coil is faulty. d) Control circuit fuses are defective.
5. Pump capacity not constant.	a) Pump inlet pressure is too low. b) Suction pipe/pump partly blocked by impurities. c) Pump draws in air.
6. Pump runs but gives no water.	a) Suction pipe/pump blocked by impurities. b) Foot or non-return valve blocked in closed position. c) Leakage in suction pipe. d) Air in suction pipe or pump. e) Motor rotates in the wrong direction.
7. Pump runs backwards when switched off.	a) Leakage in suction pipe. b) Foot or non-return valve defective. c) Foot or non-return valve blocked in open or partly open position.
8. Leakage in shaft seal.	a) Shaft seal is defective.
9. Noise.	a) Pump is cavitating. b) Pump does not rotate freely (frictional resistance) because of incorrect pump shaft position. c) Frequency converter operation: See section 8.1 <i>Frequency converter operation</i> . d) Resonance in the installation. e) Foreign bodies in the pump.

Fault	Cause
10. Pump runs constantly (applies only to pumps with automatic start/stop).	a) The stop pressure has been set too high. b) The water consumption is larger than anticipated. c) Leakage in discharge pipe. d) The direction of rotation of the pump is incorrect. e) Pipes, valves or strainer blocked by impurities. f) Pump controller, if fitted, is defective.
11. Period of operation is too long (applies only to pumps with automatic start/stop).	a) The stop pressure has been set too high. b) Pipes, valves or strainer blocked by impurities. c) Pump partly blocked or furred up. d) The water consumption is larger than anticipated. e) Leakage in discharge pipe.

13. Service



Warning

If a pump has been used for a liquid which is injurious to health or toxic, the pump will be classified as contaminated.

If Grundfos is requested to service the pump, Grundfos must be contacted with details about the pumped liquid, etc. *before* the pump is returned for service. Otherwise Grundfos can refuse to accept the pump for service.

Possible costs of returning the pump are paid by the customer.

However, any application for service (no matter to whom it may be made) must include details about the pumped liquid if the pump has been used for liquids which are injurious to health or toxic.

14. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Appendix

- GB:** Inlet pressure stated in bar relative pressure (pressure gauge value measured on the suction side of the pump)
- BG:** Относително входно налягане в бар (стойност на манометъра в смукателната страна на помпата)
- CZ:** Tlak na sání vyjádřený v barech je relativní tlak (hodnota na manometru měřená na sací straně čerpadla)
- DK:** Tilløbstrykket angivet i bar relativt tryk (manometerværdi målt på pumpens sugeside)
- DE:** Zulaufdruck in bar Relativdruck (Manometerdruck auf der Saugseite der Pumpe gemessen)
- GR:** Πίεση αναρρόφησης σε bar σχετικής πίεσης (μετρούμενη τιμή πίεσης στην πλευρά αναρρόφησης της αντλίας)
- ES:** Presión de aspiración indicada en bar como presión relativa (valor del manómetro medido en la aspiración de la bomba)
- FR:** Pression d'entrée indiquée en bar (valeur mesurée à l'aide d'un manomètre placé sur le côté aspiration de la pompe)
- HR:** Ulazni tlak u barima relativnog tlaka (manometarski tlak izmjeren na usisnoj strani crpke)
- IT:** Pressione di aspirazione indicata in bar (valore misurato con un manometro posto sul lato aspirazione della pompa)
- HU:** Hozzáfolyási nyomás a szívóoldalon bar-ban, relatív nyomás értékben (szívattyú szívóoldali nyomásmérőjén jelzett érték)
- NL:** Inlaatdruk weergegeven in bar relatieve druk (drukopnemer waarde, gemeten aan de zuigkant van de pomp)
- PL:** Ciśnienie na króćcu ssawnym pompy wyrażone w barach (mierzone manometrem na stronie ssawnej pompy)
- PT:** Pressão de entrada com a pressão relativa apresentada em bar (ponto de medida na parte de aspiração da bomba)
- RU:** Давление на входе в барах (измерения производились во всасывающей части насоса)
- RO:** Presiunea de intrare exprimată în bar ca presiune relativă (valoarea măsurată de manometru pe partea de aspirație a pompei)
- SK:** Vstupný tlak uvedený v baroch relatívneho tlaku (hodnota na manometru meraná na saciej strane čerpadla)
- SI:** Vhodni tlak v barih relativni tlak (izmerjena vrednost na sesalni strani črpalke)
- RS:** Ulazni pritisak je dat u barima relativnog pritiska (manometarska vrednost merena na usisnoj strani pumpe)
- FI:** Tulopaine ilmoitettu baareina on suhteellinen paine (painemittarin lukema mitattu pumpun imupuolella)
- SE:** Tillöppstrycket angivet i bar relativt tryck (manometervärde mätt på pumpens sugside)
- TR:** Bar olarak belirtilen nispi basınç girişi basıncı (pompanın emme kısmındaki basınç ölçü değeri)

50 Hz, 4-pole, PN 10

Pump type	p [bar]					
	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C
50 Hz, 4-pole, PN 10						
TP 150-310/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-370/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-260/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-300/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-390/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-430/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-280/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-310/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-390/4	0.1	0.1	0.1	0.8	1.4	3.1

60 Hz, 4-pole, PN 10

Pump type	p [bar]					
	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C
60 Hz, 4-pole, PN 10						
TP 150-380/4	0.7	0.9	1.4	2.2	2.7	4.4
TP 150-420/4	0.7	0.9	1.4	2.2	2.7	4.4
TP 150-490/4	0.7	0.9	1.4	2.1	2.7	4.4
TP 150-550/4	0.7	0.9	1.4	2.1	2.7	4.4
TP 200-280/4	0.4	0.6	1.1	1.8	2.4	4.1
TP 200-320/4	0.4	0.6	1.1	1.8	2.4	4.1
TP 200-350/4	0.4	0.6	1.1	1.8	2.4	4.1
TP 200-380/4	0.9	1.1	1.6	2.3	2.9	4.6
TP 200-520/4	0.8	1.0	1.5	2.3	2.8	4.5
TP 200-570/4	0.8	1.0	1.5	2.3	2.8	4.5
TP 250-450/4	1.5	1.7	2.2	2.9	3.5	5.2
TP 250-530/4	1.5	1.7	2.2	2.9	3.5	5.2
TP 250-580/4	1.4	1.6	2.1	2.9	3.4	5.1

50 Hz, 2-pole, PN 25

Pump type	p [bar]					
	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C
50 Hz, 2-pole, PN 25						
TP 100-620/2	0.2	0.4	0.9	1.6	2.2	3.9
TP 100-700/2	0.1	0.2	0.7	1.5	2.0	3.7
TP 100-820/2	0.1	0.2	0.7	1.4	2.0	3.7
TP 100-960/2	0.1	0.2	0.7	1.4	2.0	3.7
TP 1001050/2	0.1	0.1	0.6	1.3	1.9	3.6
TP 100-1180/2	0.1	0.2	0.7	1.5	2.0	3.7
TP 100-1400/2	0.1	0.2	0.7	1.5	2.0	3.7
TP 100-1530/2	0.1	0.2	0.7	1.4	2.0	3.7
TP 100-1680/2	0.1	0.1	0.6	1.4	1.9	3.6
TP 125-580/2	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-720/2	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-750/2	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-1060/2	1.6	1.8	2.3	3.1	3.6	5.3
TP 125-1160/2	1.6	1.8	2.3	3.0	3.6	5.3
TP 125-1310/2	1.6	1.8	2.3	3.0	3.6	5.3
TP 125-1500/2	1.5	1.7	2.2	3.0	3.5	5.2
TP 125-1670/2	1.5	1.7	2.2	2.9	3.5	5.2

50 Hz, 4-pole, PN 25

Pump type	p [bar]					
	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C
50 Hz, 4-pole, PN 25						
TP 100-170/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-220/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-260/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-270/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-320/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-380/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-420/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-150/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-210/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-240/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-280/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-320/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-370/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-430/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-170/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-220/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-230/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-240/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-270/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-320/4	0.1	0.1	0.1	0.8	1.4	3.1

Pump type		p [bar]				
50 Hz, 4-pole, PN 25	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C
TP 150-350/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-430/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-530/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-650/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-260/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-280/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-380/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-420/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-450/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-510/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-560/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-620/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-270/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-320/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-370/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-490/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-540/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-600/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 250-660/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 300-590/4	0.4	0.6	1.1	1.8	2.4	4.1
TP 300-670/4	0.4	0.6	1.1	1.8	2.4	4.1
TP 300-750/4	0.3	0.5	1.0	1.8	2.3	4.0
TP 400-470/4	0.1	0.3	0.8	1.6	2.1	3.8
TP 400-510/4	0.1	0.3	0.8	1.6	2.1	3.8
TP 400-540/4	0.1	0.3	0.8	1.5	2.1	3.8
TP 400-670/4	0.6	0.8	1.3	2.1	2.6	4.3
TP 400720/4	0.6	0.8	1.3	2.1	2.6	4.3
TP 400-760/4	0.6	0.8	1.3	2.0	2.6	4.3

60 Hz, 2-pole, PN 25

Pump type		p [bar]				
60 Hz, 2-pole, PN 25	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C
TP 100-1000/2	0.5	0.7	1.2	1.9	2.5	4.2
TP 100-1100/2	0.4	0.6	1.1	1.9	2.4	4.1
TP 100-1250/2	0.4	0.6	1.1	1.8	2.4	4.1
TP 100-1350/2	0.3	0.5	1.0	1.8	2.3	4.0
TP 100-1450/2	0.6	0.8	1.3	2.1	2.6	4.3
TP 100-1560/2	0.6	0.8	1.3	2.0	2.6	4.3
TP 100-1700/2	0.5	0.7	1.2	2.0	2.5	4.2
TP 100-2100/2	0.5	0.7	1.2	1.9	2.5	4.2
TP 100-2350/2	0.4	0.6	1.1	1.8	2.4	4.1

60 Hz, 4-pole, PN 25

Pump type	p [bar]					
	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C
60 Hz, 4-pole, PN 25						
TP 100-210/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-250/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-330/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-360/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-420/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-450/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-530/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 100-590/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-240/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-270/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-320/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-370/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 125-350/4	0.1	0.2	0.7	1.5	2.0	3.7
TP 125-440/4	0.1	0.2	0.7	1.5	2.0	3.7
TP 125-500/4	0.1	0.2	0.7	1.4	2.0	3.7
TP 125-570/4	0.1	0.2	0.7	1.4	2.0	3.7
TP 125-620/4	0.1	0.1	0.6	1.4	1.9	3.6
TP 150-330/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-370/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-450/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-490/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-710/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-800/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 150-930/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-350/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-370/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-520/4	0.1	0.1	0.1	0.8	1.4	3.1
TP 200-560/4	0.1	0.1	0.6	1.3	1.9	3.6
TP 200-670/4	0.2	0.4	0.9	1.7	2.2	3.9
TP 200-760/4	0.2	0.4	0.9	1.6	2.2	3.9
TP 200-840/4	0.2	0.4	0.9	1.6	2.2	3.9
TP 200-930/4	0.2	0.4	0.9	1.6	2.2	3.9
TP 250-430/4	0.3	0.5	1.0	1.7	2.3	4.0
TP 250-500/4	0.2	0.4	0.9	1.7	2.2	3.9
TP 250-550/4	0.2	0.4	0.9	1.7	2.2	3.9
TP 250-720/4	0.4	0.6	1.1	1.8	2.4	4.1
TP 250-800/4	0.4	0.6	1.1	1.8	2.4	4.1
TP 250-910/4	0.4	0.6	1.1	1.8	2.4	4.1
TP 250-1020/4	0.4	0.6	1.1	1.8	2.4	4.1

Maximum sound pressure level

Three-phase motors [kW]	50 Hz [dB(A)]		60 Hz [dB(A)]	
	2-pole	4-pole	2-pole	4-pole
5.5		58		62
7.5		58		62
11		60		64
15		60		64
18.5		61		65
22		61		65
30	69	62		66
37	69	65		69
45	72	65		69
55	74	65	78	69
75	76	69	80	73
90	76	69	80	73
110	78	70	82	74
132	78	70	82	74
160	78	70	82	74
200	78	70	82	74
250	82	73	86	77
315		73		77
355		75		
400		75		
500		75		
560		78		
630		78		

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